1 Text Mining Bibliographic Corpora

1.1 Learning objectives

After completing this chapter, you will be able to:

- Build a text corpus from OpenAlex titles and abstracts
- Apply standard preprocessing steps: tokenisation, stopword removal, stemming
- Construct a document-term matrix (DTM) and a document-feature matrix (DFM)
- Compute TF-IDF weights to identify distinctive terms
- Visualise term frequency patterns across time or groups

1.2 Setup

```
library(tidyverse)
library(openalexR)
library(quanteda)
library(quanteda.textstats)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Bibliometric metadata — titles, abstracts, and keywords — constitutes a rich but noisy text corpus. Text mining transforms this unstructured data into structured representations suitable for statistical analysis. The pipeline typically follows three stages: preprocessing, representation, and analysis.

Preprocessing converts raw text into a standardised form. Tokenisation splits text into individual words or n-grams. Lowercasing removes case variation. Stopword removal eliminates high-frequency function words ("the", "of", "and") that carry little topical information. Stemming or lemmatisation reduces words to their root forms ("computing", "computed", "computation" → "comput" or "compute"). Each step involves trade-offs: aggressive stemming can merge distinct concepts, while minimal preprocessing retains noise.

The **document-term matrix** (DTM) or **document-feature matrix** (DFM) is the fundamental representation. Rows are documents; columns are terms; cells contain counts or weights. Raw term frequencies overweight common words. **TF-IDF** (term frequency–inverse document frequency) addresses this by upweighting terms that are frequent in a document but rare across the corpus, highlighting words that distinguish one document from others.

`quanteda` [@aria2017] provides a fast, well-designed toolkit for text analysis in R. Its `corpus → tokens → dfm` pipeline integrates naturally with the tidyverse. For very large corpora, `quanteda` uses sparse matrix representations that scale to millions of documents.

Text mining complements network-based methods (**??**). Co-word networks reveal term associations; text mining provides the frequency distributions, temporal trends, and discriminative features that characterise a field's vocabulary.

1.4 Worked example

1.4.1 Building a text corpus

```
works <- fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(
    doc_id = id,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date)
  )

cat(glue("Documents with abstracts: {nrow(text_df)}\n"))
```

```
#> Documents with abstracts: 134
```

```
corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
docvars(corp, "year") <- text_df$year

cat(glue("Corpus size: {ndoc(corp)} documents\n"))
```

```
#> Corpus size: 134 documents
```

```
cat(glue("Total tokens: {sum(ntoken(corp))}\n"))
```

```
#> Total tokens: 33015
```

1.4.2 Tokenisation and preprocessing

```
toks <- tokens(corp,
               remove_punct = TRUE,
               remove_numbers = TRUE,
               remove_symbols = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("also", "however", "using", "based", "study",
                  "paper", "results", "research", "analysis")) |>
  tokens_wordstem()

cat(glue("Tokens after preprocessing: {sum(ntoken(toks))}\n"))
```

```
#> Tokens after preprocessing: 16628
```

1.4.3 Document-feature matrix

```
dfmat <- dfm(toks) |>
  dfm_trim(min_termfreq = 5, min_docfreq = 3)

cat(glue("DFM dimensions: {nrow(dfmat)} docs x {ncol(dfmat)} features\n"))
```

```
#> DFM dimensions: 134 docs x 708 features
```

```
cat(glue("Sparsity: {scales::percent(sparsity(dfmat))}\n"))
```

```
#> Sparsity: 92%
```

```
top_terms <- topfeatures(dfmat, 20) |>
  enframe(name = "term", value = "frequency") |>
  mutate(term = fct_reorder(term, frequency))

ggplot(top_terms, aes(x = frequency, y = term)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Frequency", y = NULL) +
  theme_sci()
```

1.4.4 TF-IDF weighting

```
dfmat_tfidf <- dfm_tfidf(dfmat)

tfidf_by_year <- text_df |>
  group_by(year) |>
  group_keys() |>
  pull(year) |>
  map_dfr(function(yr) {
    docs <- docvars(dfmat_tfidf, "year") == yr
    if (sum(docs) < 5) return(tibble())
    top <- topfeatures(dfmat_tfidf[docs, ], 10)
    tibble(year = yr, term = names(top), tfidf = unname(top))
  })

tfidf_by_year |>
  group_by(year) |>
  slice_max(tfidf, n = 5) |>
  gt() |>
  fmt_number(columns = tfidf, decimals = 1)
```

1.4.5 Term frequency over time

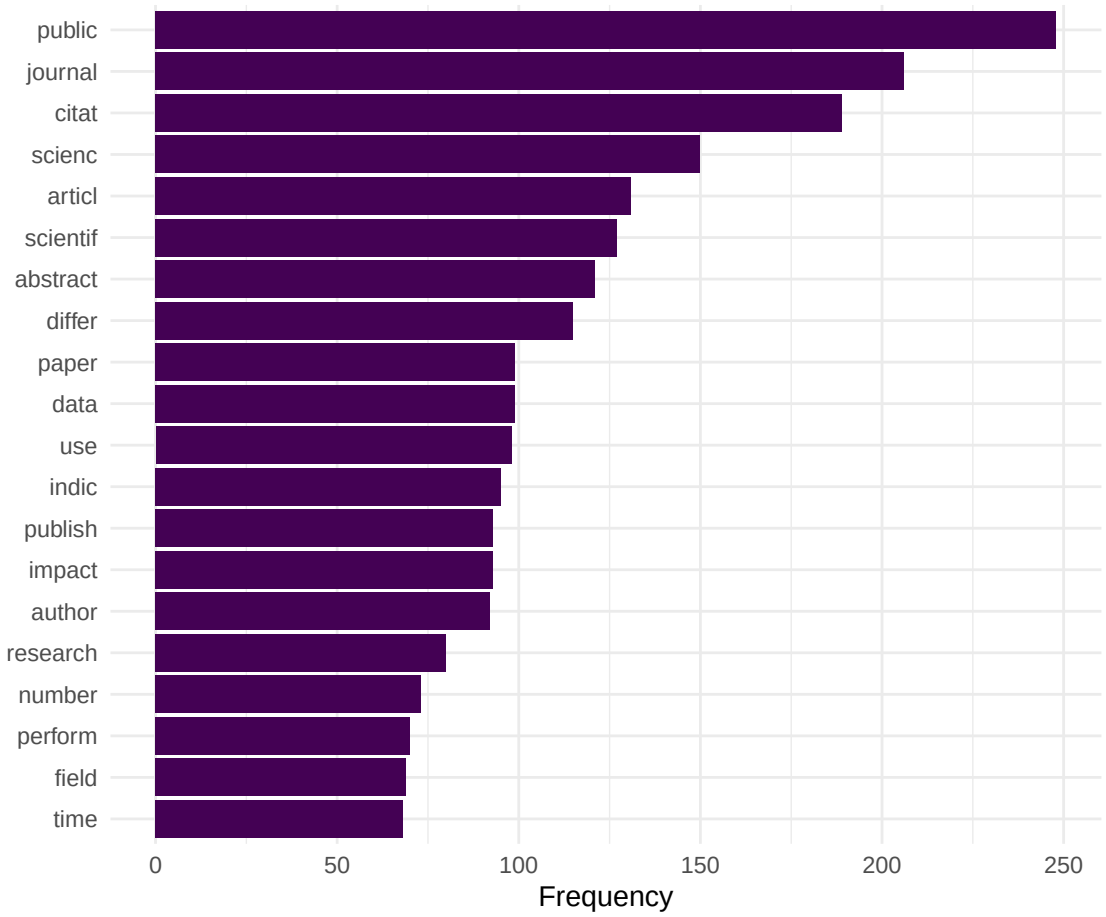


Figure 1: Top 20 most frequent terms in the Scientometrics corpus.

```r
target_terms <- c("bibliometr", "open", "network", "impact", "collabor")

freq_by_year <- map_dfr(unique(text_df$year), function(yr) {
  docs <- docvars(dfmat, "year") == yr
  if (sum(docs) < 5) return(tibble())
  freq <- colSums(dfmat[docs, ]) / sum(dfmat[docs, ])
  tibble(
    year = yr,
    term = target_terms,
    rel_freq = freq[target_terms]
  )
})

ggplot(freq_by_year, aes(x = year, y = rel_freq, colour = term)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_colour_manual(values = palette_sci(length(target_terms))) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.01)) +
  labs(x = "Year", y = "Relative frequency", colour = "Term") +
  theme_sci()
```

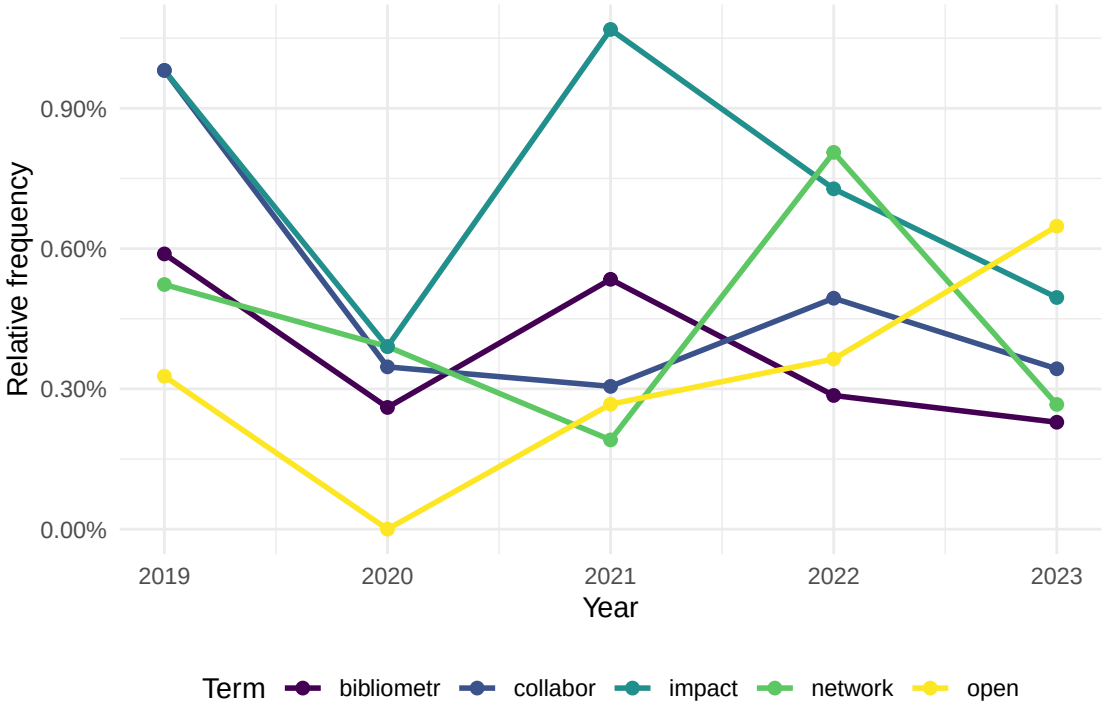


Figure 2: Relative frequency of selected terms over time.

1.5 Diagnostics and interpretation

- **Vocabulary size vs. documents**: A typical DFM has far more features than documents. Trim aggressively (minimum document frequency of 3–5) to reduce noise and computation.
- **Sparsity**: Bibliometric DFMs are typically 95–99% sparse. This is normal. Sparse matrix representations keep memory usage manageable.

- **Preprocessing sensitivity**: Results change with preprocessing choices. Always report stopword list, stemming method, and trimming thresholds.
- **Abstract availability**: Not all OpenAlex records have abstracts. Report the coverage rate and consider whether missing abstracts introduce bias (e.g., older papers, certain publishers).

1.6 Limitations and responsible use

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- **Abstracts are summaries.** They capture the main claims but miss nuance, methodology details, and negative results. Full-text analysis (**??**) provides richer data but is harder to obtain.
- **Language bias.** Text mining tools work best for English. Non-English abstracts may be poorly tokenised, incorrectly stemmed, or excluded entirely. This biases results toward Anglophone research [@visser2021].
- **Stemming conflates concepts.** "Stem" and "stemming" are merged, but so are "cell" (biology) and "cell" (spreadsheet). Lemmatisation is more precise but slower.
- **Bag-of-words ignores context.** TF-IDF and frequency-based methods treat documents as unordered collections of words. Phrase meaning ("machine learning" vs. "learning machine") is lost unless you use n-grams or embeddings (**??**).

1.8 Common pitfalls

1.9 Common pitfalls

- **Not removing domain-specific stop words.** Generic stopword lists miss terms like "study", "results", "paper" that dominate academic text without conveying topical information.
- **Applying TF-IDF before trimming.** Rare terms get extreme TF-IDF scores. Trim the DFM first, then apply TF-IDF.
- **Comparing raw frequencies across groups of different sizes.** A year with 200 papers will have higher raw counts than a year with 50. Always normalise to relative frequency.
- **Stemming titles but not abstracts (or vice versa).** Apply identical preprocessing to all text fields.

1.10 Exercises

1. **Bigram analysis.** Build a DFM using bigrams (two-word sequences) instead of unigrams. What meaningful phrases emerge that single words miss?
2. **Lexical diversity.** Compute the type-token ratio (TTR) for each publication year. Is the vocabulary growing more diverse or more standardised over time?
3. **Keyness analysis.** Use `quanteda.textstats::textstat_keyness()` to identify terms that distinguish one year from all others. What terms characterise 2023 specifically?
4. **Preprocessing sensitivity.** Run the same analysis with and without stemming. How do the top terms differ? Which version is more interpretable?

1.11 Solutions

Solutions are provided in **??**.

1.12 Further reading

- @silge2017 — Text Mining with R; comprehensive guide to tidy text analysis.
- @aria2017 — `bibliometrix` text-mining features for bibliometric data.
- @priem2022 — OpenAlex abstract availability and coverage.

1.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
 ↪  version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] quanteda.textstats_0.97.2 visNetwork_2.1.4
#>  [3] ggraph_2.2.2              tidygraph_1.3.1
#>  [5] igraph_2.3.4              quanteda_4.5.1
#>  [7] pdftools_3.9.1            arrow_26.0.0
#>  [9] bibliometrix_5.5.0        RefManageR_1.4.0
#> [11] bib2df_1.1.2.0            rcrossref_1.2.1
#> [13] gt_1.4.0                  tidytext_0.4.3
#> [15] glue_1.8.1                openalexR_3.1.2
#> [17] lubridate_1.9.5           forcats_1.0.1
#> [19] stringr_1.6.0             dplyr_1.2.1
#> [21] purrr_1.2.2               readr_2.2.0
#> [23] tidyr_1.3.2               tibble_3.3.1
#> [25] ggplot2_4.0.3             tidyverse_2.0.0
#> [27] bookdown_0.48             fs_2.1.0
#> [29] rmarkdown_2.32
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.3           RColorBrewer_1.1-3     jsonlite_2.0.0
#>   [4] magrittr_2.0.5         farver_2.1.2           vctrs_0.7.3
#>   [7] memoise_2.0.1          askpass_1.2.1          base64enc_0.1-6
#>  [10] tinytex_0.61           htmltools_0.5.9        contentanalysis_1.1.1
#>  [13] curl_8.1.0             janeaustenr_1.0.0      cellranger_1.1.0
#>  [16] htmlwidgets_1.6.4      tokenizers_0.3.0       plyr_1.8.9
#>  [19] httr2_1.3.0            plotly_4.12.1          cachem_1.1.0
```

```
#>  [22] dimensionsR_0.0.3      mime_0.13              lifecycle_1.0.5
#>  [25] pkgconfig_2.0.3        Matrix_1.7-5           R6_2.6.1
#>  [28] fastmap_1.2.0          shiny_1.14.0           digest_0.6.39
#>  [31] patchwork_1.3.2        shinycssloaders_1.1.0  rprojroot_2.1.1
#>  [34] SnowballC_0.7.1        labeling_0.4.3         urltools_1.7.3.1
#>  [37] timechange_0.4.0       httr_1.4.9             polyclip_1.10-7
#>  [40] compiler_4.6.1         here_1.0.2             bit64_4.8.6
#>  [43] withr_3.0.3            S7_0.2.2               viridis_0.6.5
#>  [46] ggforce_0.5.0          MASS_7.3-65            bibliometrixData_0.3.0
#>  [49] tools_4.6.1            otel_0.2.0             stopwords_2.3
#>  [52] zip_3.0.2              rentrez_1.2.4          httpuv_1.6.17
#>  [55] promises_1.5.0         grid_4.6.1             stringdist_0.9.17
#>  [58] generics_0.1.4         gtable_0.3.6           tzdb_0.5.0
#>  [61] rscopus_0.9.0          ca_0.72                data.table_1.18.6.1
#>  [64] hms_1.1.4              xml2_1.6.0             utf8_1.2.6
#>  [67] ggrepel_0.9.8          pillar_1.11.1          nsyllable_1.0.1
#>  [70] vroom_1.7.1            later_1.4.8            tweenr_2.0.3
#>  [73] lattice_0.22-9         bit_4.6.0              tidyselect_1.2.1
#>  [76] miniUI_0.1.2           knitr_1.52             gridExtra_2.3.1
#>  [79] crul_1.6.0             xfun_0.62              graphlayouts_1.2.5
#>  [82] DT_0.34.0              humaniformat_0.6.0     stringi_1.8.9
#>  [85] qpdf_1.4.1             yaml_2.3.12            evaluate_1.0.5
#>  [88] httpcode_0.3.0         cli_3.6.6              xtable_1.8-8
#>  [91] Rcpp_1.1.2             readxl_1.6.0           triebeard_0.4.1
#>  [94] XML_3.99-0.25          parallel_4.6.1         assertthat_0.2.1
#>  [97] pubmedR_1.0.2          viridisLite_0.4.3      scales_1.4.0
#> [100] crayon_1.5.3           openxlsx_4.2.9         rlang_1.3.0
#> [103] fastmatch_1.1-8
```

term	tfidf
2019	
econom	18.8
journal	18.3
score	17.4
doctor	16.5
peer	15.9
2020	
lis	24.7
citat	19.7
classifi	16.4
machin	14.7
covid-19	14.7
2021	
retract	41.2
book	30.5
journal	26.7
sars-cov-2	24.4
hold	21.6
2022	
journal	26.7
acknowledg	25.9
p	25.9
citat	25.3
network	24.3
2023	
mention	32.0
nobel	29.7
oa	29.7
polici	27.2
bias	23.3